Chapter 2

Periodontal disease

Angela Gilbert

CHAPTER CONTENTS

The necessary order of things 17
Susceptibility to periodontal disease 17
The concept of risk 18
Taking histories from patients with periodontal disease 18
Medical history 18
Dental history 18
Social history 19
Family history 19
Assessment of periodontal disease 19
The clinical assessment 20
The basic periodontal examination (BPE) 20
Clinical examination 21
The radiographic examination 24
Which radiographs? 24
The radiographic report 25
Making a periodontal diagnosis 26
Periodontal stability 26
Gingivitis 26
Periodontitis 26
Aggressive periodontitis 26
Periodontal treatment planning 27
The five phases of treatment planning 27
Phase 1 – Relief of pain and initial examination 27
Phase 2 – Cause-related therapy 27
Phase 3 – Re-examination 28
Phase 4 – Definitive treatment 29
Phase 5 – Maintenance or SPT 29
Periodontal considerations in relation to advanced operative dentistry 29
Margin placement 29
Temporary cover 29
Choice of advanced restorations 30
Occlusion and the periodontium 30
Summary 31

THE NECESSARY ORDER OF THINGS

There is no doubt that screening patients, diagnosing periodontal disease and then effectively treating the disease is an essential part of restorative treatment. It is crucial that successful periodontal management, demonstrated over a period of time, precedes provision of complex fixed and removable prosthodontics and implants. One analogy of this would be to consider the construction of a new building. It would make no sense to begin building walls or choosing the lighting if the foundations were not sound.

SUSCEPTIBILITY TO PERIODONTAL DISEASE

It is clear from many studies that there is considerable variation in individual susceptibility to periodontal disease. This information has shown that a relatively small percentage of the population (probably less than 15%) suffer from severe or advanced disease and are highly susceptible. A similar percentage appears to be disease resistant and

© 2011 Elsevier Ltd.

have a very low susceptibility, while the remaining 70% show some susceptibility demonstrated by moderate disease progression.

It is essential to have some understanding of an individual patient's susceptibility to periodontal disease as several things flow from this. If a patient has a high susceptibility their periodontal treatment may need to be protracted and complex if stability is to be achieved, their recall times will have to be shorter and their periodontal prognosis is reduced. For highly susceptible patients, complex fixed and removable prosthodontics can be undertaken, but *only* when both periodontal stability and adequate plaque control have been demonstrated over time – usually at least a few months. Conversely, for patients with low susceptibility, the treatment needs are usually simple, the recall times can be longer and the periodontal prognosis is good. In these patients, the decision to provide complex fixed and removable prosthodontics is much more straightforward.

Susceptibility of an individual patient is determined by relating three things:

- the age of the patient
- the plaque control
- the severity of periodontal disease

A young patient with severe disease whose plaque control is good would be regarded as having a high susceptibility whilst an elderly patient with poor plaque control and little evidence of disease would be regarded as having a low susceptibility. Susceptibility is thus a very individual concept relating to each patient. It must be appreciated that a patient's innate susceptibility cannot be changed as this is partly genetically determined. Recognition of this has led to the realization that risk recognition and control forms an essential part of periodontal management, particularly in patients with higher susceptibility.

THE CONCEPT OF RISK

It is known that the presence of risk factors will increase the chance that periodontal disease will occur. Local risk factors are essentially plaque retentive features such as restoration overhangs. These factors must be corrected before commencing periodontal treatment. Systemic risk factors may be genetic (e.g. Down's syndrome), haematological (e.g. cyclic neutropenia), behavioural (e.g. smoking, poor oral hygiene), environmental (e.g. effects of drugs, HIV), endocrine/metabolic (e.g. poorly controlled diabetes) or lifestyle (e.g. stress, poor nutrition). Factors such as smoking are modifiable whilst others, such as genetic predisposition, are not. In susceptible patients, recognition of the risks and control of the modifiable risk factors is essential if periodontal therapy is to be successful. This is particularly true if provision of more complex fixed and removable prosthodontics is envisaged.

Risk recognition begins at the stage of taking thorough histories from patients.

TAKING HISTORIES FROM PATIENTS WITH PERIODONTAL DISEASE

Taking thorough histories is an essential part of a patient's oral health care and is particularly relevant to periodontal therapy. This takes time but should not be ignored or truncated as the information derived is essential if appropriate decisions regarding treatment planning, provision of periodontal therapy and maintenance are to be made. This again is particularly important when periodontal management precedes and is integrated into treatment plans that involve complex fixed and removable prosthodontics.

There are four main histories to be taken. These are Medical, Dental, Social and Family histories, and aspects of relevance to periodontal disease are highlighted here.

Medical history

This serves three main functions:

1. It may explain, at least in part, the severity and extent of periodontal disease encountered. An example would be the increased disease which may be seen in some patients with poorly controlled diabetes. If control is not improved, the chance of producing a successful periodontal treatment outcome is reduced, thus making the later provision of complex restorative work unlikely.
2. It may alert the clinician to precautions which should be taken to safeguard the patient. This may include provision of steroid cover or highlight the need for treatment in a specialist or hospital setting. These complications may also make the provision of complex restorative work less likely.
3. It may alert the clinician to potential risks posed by a patient to staff and subsequent patients in the surgery. Standard cross-infection procedures should be operated for all patients but it would be wrong, for example, to use an ultrasonic scaler in a patient carrying a known infective risk. Ultimately, however, provided periodontal stability has been achieved, there should be no reason not to proceed with complex restorative work if this was the desired outcome.

Dental history

It is clear that patients who attend the dentist irregularly are not candidates for advanced restorative treatment plans. In addition to the standard parts of a dental history, particular aspects that should be assessed in relation to periodontal health are as follows:

- *Plaque control regime used*. If there is active periodontal disease in the mouth, plaque control is, by definition, 'inadequate' and must be modified. It is usually very straightforward to recommend appropriate changes to the techniques and frequencies employed; however, it must also be recognized that plaque will potentially cause caries. This means that even if plaque control is sufficient to control the periodontal condition, it may not be sufficient to prevent damage to the teeth themselves. If plaque control is not subsequently improved, the provision of complex restorative treatment would not be indicated.
- *Wearing a removable appliance*. Use of an appliance such as a denture is relevant as numerous studies have shown that this increases plaque retention, which in turn may be sufficient to take the plaque levels above the patient's individual disease threshold. Adequate denture hygiene and advice to remove the appliance at night are essential.
- *A previous history of periodontal disease*. This is the best indicator of future risk because the innate susceptibility has not changed. As stated previously, provided periodontal stability is demonstrated over time, and provided the plaque regime is adequate to control both caries and periodontal disease, provision of complex operative care should not be withheld. It must be recognized, however, that the patient is not 'cured' and it is worthwhile considering that the disease may reactivate.
- *A history of previous periodontal treatment*. This may indicate both the patient's attitude to treatment and the degree of difficulty encountered in treating their previous disease. If it has proven difficult to stabilize the periodontal condition in the past (repeated root surface instrumentation, periodontal surgery, use of antimicrobials, etc.), this suggests a high susceptibility and increased chance of subsequent periodontal breakdown. If complex restorative work was being considered in such circumstances it would be wise to delay this until periodontal stability had been demonstrated over several months at least. Short recall times would also be needed for a substantial period after placement.

Social history

There are three particular aspects of the social history that are relevant to periodontal disease. The single most important factor is a history of smoking. Recent research suggests that smoking accounts for up to 50% of cases of chronic periodontitis and it has been established that the incidence and severity of periodontal disease increase with increasing tobacco exposure. Tobacco exposure should be quantified as 'pack years' and this is calculated as the number of packs of 20 cigarettes smoked per day multiplied by the number of years for which the patient has smoked. For example, smoking 20 cigarettes per day for 10 years would equate to 10 pack years. This provides a better indicator of total nicotine exposure than knowing that someone smokes 20 cigarettes per day but without knowing for how long they have had the habit. Some patients do not smoke standard cigarettes and for them it is possible to convert to 'pack years' by assuming that 1 g of pipe tobacco is equivalent to one cigarette, one small cigar is equivalent to 3 cigarettes and 1 standard cigar is equivalent to 5 cigarettes. The impact of 'pack years' is also relevant to implant assessment as it is widely accepted that smoking reduces the success of osseointegration.

The other two factors that are thought to influence periodontal disease, but to a lesser extent, are alcohol consumption and stress. Several sources, including the Third National Health and Nutrition Examination Survey (NHANES), United States, have found a moderate but consistent dose-dependent relationship between alcohol consumption and periodontal disease. The recommended maximum number of units of alcohol per week for a woman is 14 and for a man it is 21. In times of stress, amongst other things, a patient's oral hygiene and oral health care may deteriorate, their nutrition may suffer, their salivary flow decreases, immune function is depressed, and plaque becomes stickier and thicker. Stress is not easy to measure, however, and although regarded as a risk factor, its single impact on dental care is difficult to determine.

Family history

The genetic links associated with aggressive periodontitis (AgP) are much stronger than with chronic periodontitis. It is important to know if there is a family history of periodontal disease, particularly involving early loss of teeth, as this may be significant in making the periodontal diagnosis. AgP, as the name suggests, is characterized by a very rapid loss of periodontal support and referral to a specialist periodontist is often indicated. Clearly, there should be no provision of complex restorative work until the periodontal condition is stabilized. Fortunately, many AgP cases 'burn out' and when stability is subsequently demonstrated over time, advanced restorative care can be considered.

It should also be noted that siblings or children of AgP patients should be regularly screened for the disease.

ASSESSMENT OF PERIODONTAL DISEASE

A clinical examination and where necessary sensitivity testing, will inform the clinician as to whether radiographic examination is required and if so which views are required in order to produce a diagnosis and treatment plan.

The clinical assessment

The basic periodontal examination (BPE)

The BPE (Table 2.1) is a screening tool to enable practitioners to determine whether or not their patient has significant periodontal disease. This examination should be used to screen:

- all new patients
- all those patients who have not had a periodontal examination of any kind in the past year.

The BPE can only be performed using a WHO probe and the probing force applied should be 20–25 g. There are two common variants of the WHO probe in use: the WHO-E (epidemiological) type probe and the WHO-C (clinical) type probe (Figure 2.1). The key elements to note on each are a ball-shaped probe tip of diameter 0.5 mm and a coloured band extending from 3.5 to 5.5 mm. The WHO-C probe has, in addition, a second coloured band extending from 8.5 to 11.5 mm.

To carry out a BPE, the dentition is divided into sextants (first premolar to second molar and canine to canine). The probe tip is gently placed in the base of the gingival crevice/pocket and 'walked' around all the teeth in the entire sextant. One code is assigned per sextant and this is the highest encountered anywhere within it. Table 2.1 shows the criteria used for assigning the BPE codes and the corresponding management for each sextant.

Third molars are not included in the BPE and at least two teeth must be present in a sextant for it to be coded. If there is only one tooth in a sextant this is included in the adjacent sextant and if there are no teeth present in a sextant this is denoted on the completed chart as 'X'.

An example of a completed BPE chart is shown below:

X	4	3
*	1	2

Table 2.1 BPE codes, together with the description, a clinical example and appropriate management

BPE CODE	DESCRIPTION	CLINICAL PICTURE	MANAGEMENT
0	The coloured band (3.5–5.5 mm) is completely visible and there is *no* bleeding on probing from the base of the pocket		No treatment required
1	The coloured band (3.5–5.5 mm) is completely visible plus there *is* bleeding on probing from the base of the pocket		Oral hygiene instruction (OHI)
2	The coloured band (3.5–5.5 mm) is completely visible plus either calculus (supra or subgingival) or a defective restoration is present		OHI plus removal of calculus ± correction of plaque retentive margins of restorations

(*Continued*)

Table 2.1 BPE codes, together with the description, a clinical example and appropriate management—cont'd

BPE CODE	DESCRIPTION	CLINICAL PICTURE	MANAGEMENT
3	The coloured band (3.5–5.5 mm) is only partly visible		Full charts required at start and finish of treatment. OHI plus calculus removal, correction of defective margins of restorations and root surface instrumentation (RSI) as required
4	The coloured band (3.5–5.5 mm) is completely hidden		
*	Loss of attachment of ≥7 mm		Full charts required at start and finish of treatment, including note of furcation involvements and any other relevant clinical findings OHI plus calculus removal, restoration margin correction and RSI as appropriate. Surgery may be needed and consideration should be given to referring the patient to a periodontal specialist
*	Furcation involvement		

Should significant disease be found (BPE Codes 3, 4 or *), a more detailed periodontal, clinical and possibly radiographic examination is required (see later).

As with any system, the BPE has both advantages and disadvantages (Table 2.2) and the reader should be aware of these if the system is to be correctly applied and interpreted.

Clinical examination

For all patients who have BPE codes 3, 4 or *, a full periodontal examination is required. This needs to be performed at all sites in the mouth as there may be variation in the severity of disease at different sites and at different times. This is because periodontal disease is

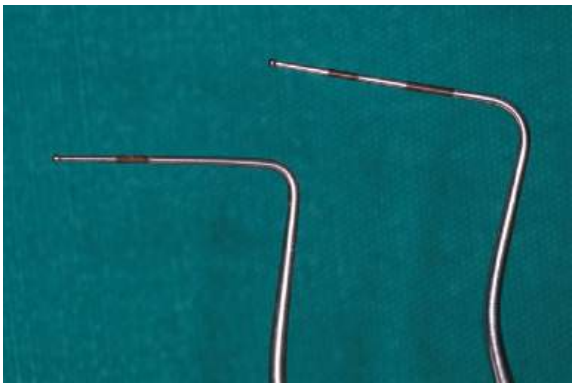


Figure 2.1 WHO-E (epidemiological) type probe (left) and the WHO-C (clinical) type probe (right).

recognized as being both site specific and episodic in nature.

The five essential clinical assessments required to determine extent and severity of periodontal disease are as follows:

- *Probing* (*pocket*) *depth*. This is measured from the gingival margin to the base of the pocket. Pocket depths deeper than 3 mm are regarded as being significant as patients cannot efficiently clean them. Reduction of pockets to this depth is thus one of the principal aims of therapy.
- *Loss of attachment*. This is measured from the amelo-cemental junction (ACJ) to the base of the pocket. This represents the sum of all the episodes of disease which have taken place at a particular site since the tooth erupted, though it does not indicate how many episodes of disease have occurred or when they occurred. A completed pocket/loss of attachment chart (Figure 2.2) is particularly useful in determining the severity of periodontitis and aiding treatment planning in terms of sites that may require root surface instrumentation (RSI). The chart will also differentiate between true and false pockets and make identification of sites with recession/tissue shrinkage and/or gingival overgrowth clear.
- *Lack of bleeding on probing* (*BOP*) *from the base of the pocket*. This indicates a lack of disease activity at a site at a particular time. Virtually all sites not showing BOP will not go on to lose attachment. Previously, sites which did show BOP were regarded as a positive indicator of disease activity, but unfortunately only

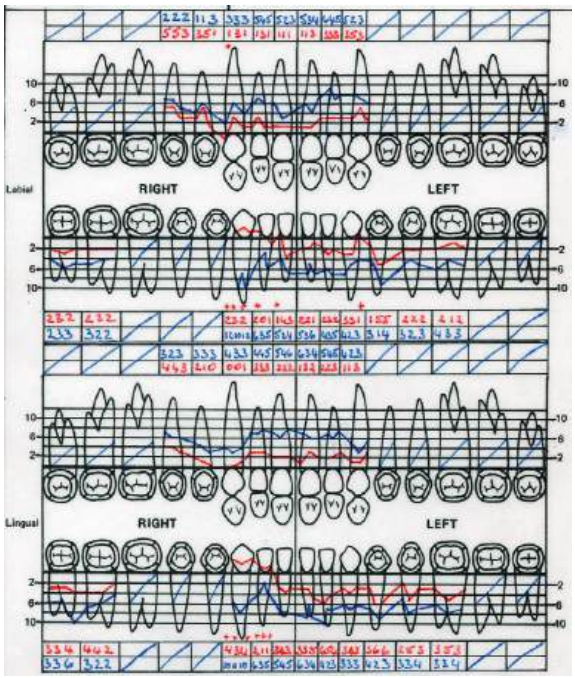


Figure 2.2 Completed pocket/loss of attachment chart.

Table 2.2 Advantages and disadvantages of the BPE

ADVANTAGES	DISADVANTAGES
• It is a screening system which detects individuals requiring a more detailed periodontal assessment • It is recognized internationally • It is quick and easy to carry out and the required equipment is inexpensive • It encourages examination of the periodontium in general dental practice • It summarizes the periodontium in a readily communicable form • It gives an indication of treatment requirements and who should treat the patient – should the patient be referred to a specialist?	• It lacks detail within sextants • It cannot be used to monitor disease • It requires a special probe • It does not distinguish between true and false pockets. • It lacks information on disease activity. bleeding on probing (BOP) is only mentioned in Code 0 (where it is not present) and in Code 1 (where it is present) • It lacks detailed information on loss of attachment (LOA). LOA is mentioned in Code * alone • It lacks detailed information about furcation involvement

30% of sites showing BOP will go on to lose attachment. Using lack of bleeding is thus a much more robust assessment. A bleeding chart (Figure 2.3) may be used to illustrate this. Both gingivitis and periodontitis can be said to occur only if bleeding is present. This chart is particularly useful in determining the distribution of disease, i.e. whether it is localized or generalized (see later in section on diagnosis).

- *Furcation involvement*. This should be assessed in all multirooted teeth. Involvement is significant because affected teeth have a reduced prognosis. This is partly because these sites are often extremely difficult for both dentist and patient to clean and partly because the affected teeth may lose vitality. Curved furcation probes (Figure 2.4) are useful to facilitate this part of the examination (Figure 2.5). When reporting furcation involvement, the clinician should note specifically which tooth is affected, which furcation is affected and the severity of the involvement at affected sites. There are several furcation severity grading systems in use but the one suggested by Hamp et al. (1975) shown in Table 2.3 is particularly helpful. Lastly, every tooth with furcation involvement should be sensitivity tested. This should be done with cold stimulation and an electric pulp tester to minimize the possibility of obtaining both false-positive and false-negative results (see Chapter 3). If a tooth has a true combined (perio-endo) lesion, the endodontic treatment must be carried out first. If furcations are exposed in teeth that are planned for crowns and bridges, special care needs to be taken in tooth preparation and restoration contour to avoid positive ledges in the furcation area.
- *Tooth mobility*. There are several dental causes of tooth mobility of which periodontal disease is only one. Other common dental causes include occlusal trauma and periradicular disease. It is essential to establish both the cause(s) of the observed mobility and its severity. Grading the degree of mobility should be done using two instrument handles to check for movement and it is important to use a suitable index to reduce the subjectivity of the assessment. One such index is Miller's Index which is described

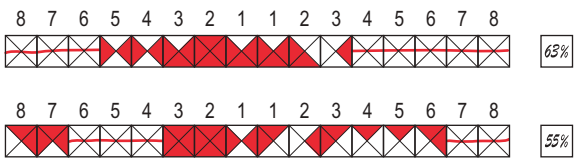


Figure 2.3 Bleeding chart. As for the plaque chart, the circumference of the gingiva around the tooth is divided into quarters, and the site(s) where bleeding occurs recorded. A percentage bleeding score can then be calculated.

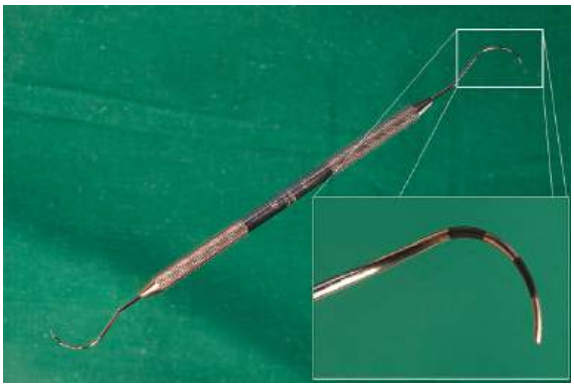


Figure 2.4 Furcation probe (main image) with head magnified (inset).



Figure 2.5 Furcation probe demonstrating buccal furcation in the lower right second molar tooth.

Table 2.3 Furcation classification described by Hamp et al. (1975)

FURCATION CLASSIFICATION	DESCRIPTION
1	Horizontal loss of periodontal support $\leq 1/3$ of the width of the tooth
2	Horizontal loss of periodontal support $> 1/3$ but not encompassing the total width of the tooth
3	Horizontal 'through and through' destruction of the periodontal tissues in the furcation area

Based on Hamp S, Nyman S, Lindhe J. Periodontal treatment of multirooted teeth. Results after 5 years. J Clin Periodontol 1975;2:126–135.

in Table 2.4. It is important to note that failure to elucidate the reason(s) for tooth mobility precludes effective treatment. The distribution and severity of tooth mobility may also be recorded on a chart (Figure 2.6).

It will be noted that an assessment of plaque control is not included in the assessment of disease. This is because there is no direct link between the amount of plaque and the presence, severity and activity of periodontal disease (see above for discussion of susceptibility). It will of course be assessed separately after the above examinations are completed and a plaque chart may be useful in this regard (Figure 2.7). This chart records the presence of plaque only and makes no attempt to quantify it. If plaque control is inadequate (i.e. there is active periodontal disease and/or caries), this chart identifies the sites the patient is not cleaning and appropriate targeted oral hygiene instruction may be given.

There may or may not be an obvious correlation between sites with bleeding and sites with plaque in the same mouth. Although correlation will depend upon the innate susceptibility of the patient, it must be recognized that plaque varies greatly in composition and thus pathogenicity. It is also true that plaque scores taken on the day of presentation may not be representative of the usual plaque control exercised by patients who may clean unusually well before their appointment. However, even with all these considerations it is possible to gain some insight into the susceptibility by comparison of the two charts. This is especially the case if the plaque scores are high and the bleeding scores are low, implying a low susceptibility to disease.

In addition to these parameters it is often useful to know whether the patient is suffering from problems with hypersensitivity (see Chapter 4 for the management of dentine sensitivity), inadequate function or an aesthetic problem such as tooth migration.

Completion of these examinations establishes the presence, location and severity of any periodontal disease, and it is only at this juncture that the clinician can establish whether a radiographic examination is required and, if so, which radiographs are appropriate.

Table 2.4 Miller's index for tooth mobility

MOBILITY CLASSIFICATION	DESCRIPTION
0	Horizontal movement up to 0.2 mm
1	Horizontal movement of >0.2 mm but ≤1 mm
2	Horizontal movement of >1 mm
3	Movement in both horizontal and vertical planes

Based on Miller SC. Textbook of periodontia. Philadelphia: Blakiston, 1938.



Figure 2.6 Mobility chart.

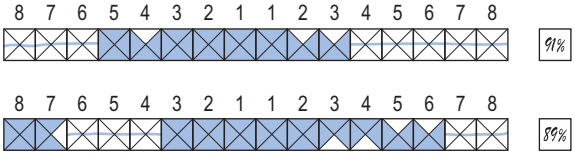


Figure 2.7 Plaque chart. Each tooth is divided into quadrants and the presence of plaque recorded in each quadrant. The percentage surface coverage in plaque can be calculated.

The radiographic examination

A radiographic examination, if required, is a useful adjunct to the clinical examination and not an alternative. At all times a balance is needed between the diagnostic yield from the radiograph and the amount of radiation to which the patient is exposed. It is mandatory that the radiation exposure is kept to the minimum for the diagnostic purpose.

Which radiographs?

There are published guidelines about which radiographs to use in particular situations, and although these are useful, there are often individual considerations to ponder. It is important not to take radiographs if there is no definite clinical indication to do so. An example of this would be in cases of chronic gingivitis – irrespective of whether the condition is localized or generalized, there will be no useful diagnostic yield.

In periodontology three main radiograph types are employed: dental panoramic tomograms (DPTs), peri-apicals, and both vertical and horizontal bitewings. The choice depends on the results of the individual clinical examination. Horizontal bitewing radiographs have very limited use in periodontology as furcation areas and even the alveolar bone crest may not be visible, depending upon the severity of the bone loss. Vertical bitewings have

their advocates as a greater proportion of each tooth is seen, but relatively few teeth are shown per radiograph and the apical areas are not consistently visible.

The quality of DPT radiographs has greatly improved in recent years and these can give an excellent overview of the jaws. However, the detail is not as good as that produced by peri-apicals and this is particularly true in the anterior region where there is often superimposition of the cervical spine (Figure 2.8). If fine detail of the entire dentition is required it is often wise to take peri-apical radiographs at the outset. This would be appropriate if there was severe periodontal disease and/or the teeth were heavily restored. In such cases it is necessary to see apical areas, bone support and furcation areas clearly. It may be tempting to take a DPT first and supplement this later with additional peri-apicals; however, if several such peri-apicals are taken, the total radiation associated may exceed that which would have been required for full mouth peri-apicals at the outset (Figure 2.9). If detailed examination of specific areas only is required, then only the appropriate and minimum number of radiographs should be taken.

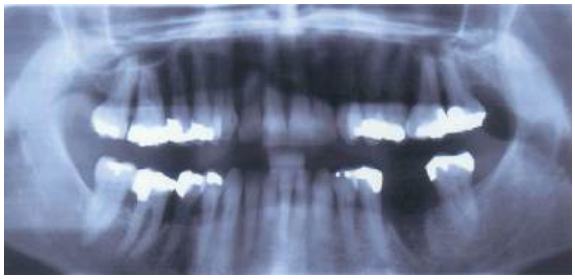


Figure 2.8 DPT radiograph of patient with moderate to severe chronic periodontitis. Note the image in the upper arch, especially in the anterior region, is unclear.

The radiographic report

Completing the radiographic report as soon as possible is an essential discipline if the diagnostic yield is to be maximized. The information gleaned invaluably informs diagnosis and treatment planning. Of particular relevance to periodontal disease, a note should be made of the presence of restoration overhangs/deficiencies that will make plaque removal impossible; such defects must be corrected if periodontal therapy is to be successful. Radiographs should be checked for the presence of calculus 'wings' although these are seen only on proximal surfaces and only if the calculus is sufficiently mineralized. Dark triangular 'arrowhead' lesions seen between the roots of multirooted teeth suggest the presence of furcation involvement though this must always be confirmed clinically. Periradicular lesions should also be noted.

Presence of periodontal bone loss can only be evaluated by looking at interproximal bone. This is because the teeth are so radio-opaque that buccal or lingual bone overlying them is obscured. Bone loss should be described under three headings: the distribution (whether localized or generalized), the pattern (whether horizontal or vertical) and the severity. Bone loss is localized if up to and including 30% of sites are affected; it is generalized if more than 30% of sites are affected. There are several methods of assessing severity but it is probably best to look at the percentage bone loss in relation to the root length.

The radiographic findings are critical not only from a periodontal point of view but also give an idea of the long-term prognosis of the tooth/teeth and suitability for use in more complex treatment plans involving indirect restorations. This having been said, it is important to remember that radiographs have distinct limitations. They give no information about soft tissues and so are not helpful when assessing pocket depth, loss of attachment, disease activity (lack of bleeding on probing) or tooth mobility. They also give no indication of the duration

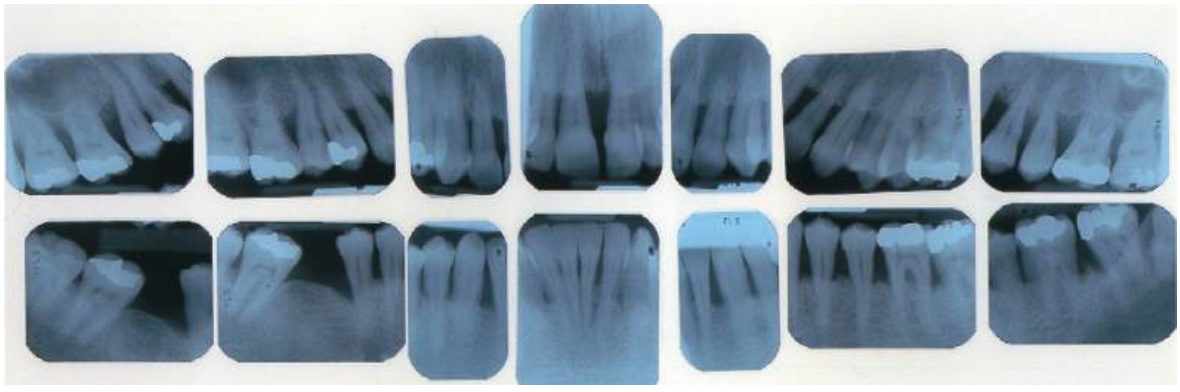


Figure 2.9 Full mouth periapical radiographs of a young patient with aggressive periodontitis. More detail is provided for each tooth compared to that obtained from a DPT. The radiation involved is justified due to the severity of the periodontal disease.

of the periodontal disease, the number of periodontal disease episodes or the timing of these episodes.

MAKING A PERIODONTAL DIAGNOSIS

A periodontal diagnosis should be formulated for all new patients, and also at all scheduled recall and any unscheduled appointments should an acute periodontal condition arise. An individual diagnosis is useful in its own right on a given day as well as being useful in a comparative way over a period of time. If a patient originally had a 'localized moderate chronic periodontitis' but over a time this had changed to become a 'generalized severe chronic periodontitis', this would highlight that the disease had become both more widespread and more severe. Again, if such changes took place over a known period of time, the rate of change would also be known. If there had been a high rate of change, advanced operative procedures would be contraindicated. This would affect treatment planning and may prompt further investigations should an underlying pathogenesis such as a systemic disease be suspected.

Periodontal diagnosis is often regarded as being a complex and difficult subject, but the reality is that there are really only three basic periodontal diagnoses. These can be modified and applied to any given situation. There has been general agreement on the classification of periodontal disease using the 1999 World Workshop system and the reader is advised to consult this for reference. The three basic diagnoses are as follows.

Periodontal stability

This is the diagnosis when two conditions are satisfied: (1) there is a lack of bleeding upon probing, implying a lack of disease activity; and (2) there are no pockets greater than 3 mm in depth (pockets shallow enough for the patient to maintain). This is the desirable situation and is more likely in patients with a low susceptibility to periodontal disease. It is also seen in successfully treated periodontal patients with higher susceptibility but it must be remembered that such patients are not cured and that recurrence of disease is possible. This is the ideal situation in which to begin planned complex restorative work provided the stability is demonstrable over a period of time.

Gingivitis

This is the diagnosis when there is bleeding on probing and there is no loss of attachment. If this is the case, two descriptors need to be applied to complete the diagnosis – namely whether the disease is localized (up to and including 30% of sites affected) or generalized (greater than 30% of sites affected). It is also important to name the type of gingivitis present. By far the most common is chronic gingivitis though the reader should consult the 1999 Classification of Periodontal Disease to see the other possibilities.

A diagnosis can then be constructed (e.g. generalized chronic gingivitis). Should localized gingivitis be present, the sites affected should be highlighted (e.g. localized chronic gingivitis affecting lower incisors and canines). There is no need to apply any description of severity of gingivitis as there is no loss of attachment. Gingivitis should not be ignored, however, as it is the precursor for periodontitis in susceptible patients which is irreversible.

Periodontitis

This is the diagnosis when two conditions are satisfied: (1) there is bleeding on probing; and (2) there is loss of attachment. If this is the case, three descriptors need to be applied to complete the diagnosis. These are:

- whether the disease is localized or generalized (assessed in the same way as for gingivitis)
- the severity of the periodontitis (based on loss of attachment (LOA); a LOA of 1–2 mm is defined as 'slight or mild', LOA of 3–4 mm is defined as 'moderate' and LOA of 5+ mm is defined as 'severe')
- the type of periodontitis (while chronic periodontitis is the most common type of periodontitis, the reader is again directed to the 1999 classification system to see other recognized conditions, including aggressive periodontitis).

In a similar way to gingivitis, a diagnosis can then be constructed (e.g. generalized moderate chronic periodontitis). Further refinements can be made to describe any given situation accurately (e.g. generalized moderate chronic periodontitis with localized severe chronic periodontitis affecting lower incisors and canines).

Aggressive periodontitis

It is worth making special mention of this particular form of periodontitis. This is not because it is very common (it affects only ~0.1–0.2% of caucasians and 2.6% of black populations.) but because there is often a strong family history in this disease and, if present, it can cause rapid irreversible loss of periodontal support leading ultimately, in some cases, to loss of teeth. It must be diagnosed as early as possible and treated effectively. There are recognized radiographic and clinical features of the condition (some of which are illustrated in Figures 2.9 and 2.10, respectively) and clinicians should be familiar with these (see Clerehugh 2004). In general, prompt referral to a specialist periodontist follows diagnosis of this condition. It is likely that treatment will involve both physical hands-on



Figure 2.10 Clinical images of the patient seen in Figure 2.9. Note that plaque control is good. There is migration of both the upper right central incisor and canine. It would be easy to greatly underestimate the severity of periodontal destruction using clinical visual assessment alone.

periodontal therapy (with or without surgical access) plus prescription of systemic antimicrobials. In the past, the drug choice would have been oral tetracycline 250 mg four times per day for 2–3 weeks. Unfortunately, increasing bacterial resistance means that this is no longer as effective in some patients as it once was and a new drug regimen has been recommended. This is oral metronidazole 400 mg *plus* oral amoxicillin 250 mg, both three times per day for 1 week.

PERIODONTAL TREATMENT PLANNING

Periodontal treatment planning is straightforward and the following section gives a logical outline for the clinician to follow. It can be infinitely adapted to the requirements of individual patients, with inclusion of different permutations of the elements being necessary in some cases but not in others. There is no single complete blueprint for periodontal therapy that can be applied indiscriminately to all patients as the overall disease experienced, susceptibility and attendant risk factors vary between individuals.

The five phases of treatment planning

It is useful to consider the complete treatment plan within this framework. Whilst in this chapter there is focus on periodontal treatment planning, the same principles can be applied to other conditions such as caries and tooth wear, and a multidisciplinary treatment plan can be constructed.

Phase 1 – Relief of pain and initial examination

It is relatively unusual for periodontal disease to cause pain, but should this be the case investigation, diagnosis and relief of this is an absolute priority. The cause of the discomfort may or may not be periodontal in origin. At this point, history taking and examination of the periodontium along the lines discussed earlier would be undertaken so that a provisional diagnosis and treatment plan can be formulated.

Phase 2 – Cause-related therapy

The primary purpose of this stage is essentially to control risks. If the radiographic examination revealed caries, presence of overhangs, suggestion of furcation involvement,

presence of periradicular areas, etc., these should be confirmed clinically and appropriate treatment prescribed. Treatment elements here will include whichever of the following are relevant to the patient:

- Clinical evaluation of caries, plaque retentive factors, possible furcation involvements and sensitivity testing of appropriate teeth
- Oral hygiene instruction. The plaque chart will show whether smooth surface (tooth brushing) and/or interproximal instruction is required. Where possible, interproximal cleaning is best achieved by TePe brushes of the correct size (Figure 2.11)
- Removal of plaque-retentive factors such as overhangs
- Smoking cessation advice
- Alcohol reduction advice
- Extraction of teeth of hopeless prognosis
- Scaling and polishing of teeth
- Root surface instrumentation (RSI) of sites with true pockets $\geq$4 mm in depth which also show BOP and where the patient has achieved excellent plaque control
- Dietary advice. This may be required if caries or erosion has been found in the mouth. A diet diary may be useful to highlight both sugar and acid sources and consumption frequency (see Chapters 1 and 4)
- Stabilization of caries if necessary.

Phase 3 – Re-examination

This stage is absolutely essential and should be timed for a minimum of 6–8 weeks after the last RSI has taken place. This time is the minimum required for healing to occur. It is hoped that completion of Phase 2 will achieve control of the periodontal disease but this is not certain. Clinical re-examination at this time (preferably with completion of new periodontal charts if they were used at the beginning of therapy) will reveal whether disease is controlled or not.



Figure 2.11 TePe brushes for interdental cleaning.

It is also at this stage that for the first time it will be possible to assess both the tissue response to treatment and patient compliance. This aspect of care is important. The time needed for re-examination is necessary to determine whether periodontal stability has been achieved or not. Complex prosthodontic treatment cannot be considered until it has.

Three outcomes are possible at this time:

- *Clear treatment success*. This is the outcome if periodontal stability has been achieved. This is the desirable outcome but it must be remembered that the patient is not 'cured'. Patients with this outcome progress directly to Phase 5 – Maintenance or supportive periodontal therapy (SPT). It is only when this outcome has been achieved should more complex operative intervention such as crowns and bridges be considered, and even then only when this stability is maintained successfully for some time – perhaps up to 6 months or even more.
- *Clear treatment failure*. This will be apparent as persistence of BOP and continued presence of pockets too deep for the patient to maintain. At this point it is essential to establish why failure has occurred and whether the reason(s) can be controlled. If this is not established, further treatment is likely to fail. Although the most common cause for failure is inadequate plaque control, there are others that include inadequate debridement, incorrect diagnosis of the condition (e.g. presence of an undetected endodontic lesion or root fracture) or the patient having a refractory response to treatment (fortunately a very rare scenario).

  If it is possible to correct the reason(s) for failure, the patient progresses to Phase 4 – Definitive treatment, with reasonable hope that a positive outcome may be achieved. If, however, it is not possible to correct the reason(s) for failure, thus leading to persistence of disease (e.g. the patient is unable or unwilling to achieve adequate plaque control), palliative care is recommended. The aim of palliative care is to keep the patient comfortable and to maintain a functioning dentition for as long as reasonably possible. Treatment will be simple – often scaling and polishing every 3 months. It is essential that the patient understands this situation and details must be entered in the patient records to avoid any future medico-legal complications should extractions become necessary. Patients on palliative care are not good candidates for provision of complicated restorative work, partly as the foundation of the dentition is compromised and partly as placement of any structures which further compromise plaque removal is likely to hasten the demise of the remaining dentition.
- *Intermediate result*. In this case there is no BOP (and thus no disease activity) but there are still some

periodontal pockets which are too deep for the patient to maintain. This outcome is common and may represent sites which are healing more slowly. As there is no disease activity, further RSI at these sites would be contraindicated. Equally, however, deep pockets which cannot be maintained by patients may become active again. At this stage RSI must *not* be undertaken. The way to deal with this apparent conundrum is to give any necessary oral hygiene instruction and perform any required scaling and polishing, and then review the patient again in approximately 1 month. At this time healing may have progressed and the patient can move into the maintenance phase (Phase 5); alternatively, the involved sites may have become active once more (BOP is present) and the probing depths are still too deep for the patient to maintain themselves. If this latter situation pertains, the patient should be managed as for 'clear failure' as described above. As always, the clinician should not proceed with advanced restorative work until stability has been achieved and maintained for sufficient time. Premature complex treatment may result in reactivation of the periodontal disease and ultimate failure of the overall treatment. Patience is rewarded!

Phase 4 – Definitive treatment

This often includes re-root surface instrumentation of sites. Should this prove ineffective, and the conditions for surgical intervention are satisfied, periodontal surgery may be undertaken. The planning of the type and extent of the surgery cannot be judged until this stage has been reached.

Only once stability has been achieved does the patient progress to Phase 5.

Phase 5 – Maintenance or SPT

This is basically a recall system that patients only commence once periodontal stability has been achieved and that aims to keep them in this stable condition. There are many factors to consider when setting the recall times but in general they will start being short (certainly no more than a couple of months if RSI was required as part of the treatment) and will lengthen only when stability is demonstrated over time.

Reaching Phase 5 does not mean that the patient remains in it in perpetuity. Should disease recurrence occur, the patient will have to re-enter the earlier phases of treatment and reach Phase 5 again only once stability has been regained. Patients thus enter, leave and re-enter Phase 5 as dictated by their disease activity. Advanced operative procedures should not be undertaken until Phase 5 has been reached and maintained for an acceptable period.

PERIODONTAL CONSIDERATIONS IN RELATION TO ADVANCED OPERATIVE DENTISTRY

Should active periodontal disease recur during or after provision of advanced operative dentistry, this must be recognized as early as possible. Failure to do this may result in rapid deterioration of the periodontal condition that may become progressively more difficult to treat. It is essential that the patient is regularly examined during this time and, if necessary, a rapid referral for advice/treatment to a specialist periodontist may be warranted.

Margin placement

It is much easier to get good impressions and thus well-fitting fixed and/or removable prosthodontics if the periodontium is healthy. The gingival margins should be stable and the tissue of good quality. This means that there should be no or limited bleeding when impressions are taken for indirect restorations such as crowns. Limited gingival retraction may be required and methods to achieve this are detailed in Chapter 13. If the tissue is healthy there should be little or no need for gingival electrosurgery.

Margin placement is of great importance and there is no doubt that, where possible, it should be supragingival. It may be tempting to place margins on visible labial surfaces subgingivally in an attempt to maximize aesthetics. This is ultimately futile as this complicates and often compromises the impression-taking process, makes checking and fitting of margins more difficult, and often results in marginal plaque accumulation. This will result in some degree of gingival inflammation that in turn results in an aesthetic deficit and often activation of ongoing periodontal disease. Gingival swelling or shrinkage may then result and, in the latter case, the margin placed subgingivally may become supragingival. It must be better to place restoration margins on visible labial surfaces just within the gingival crevice.

Temporary cover

Well constructed temporary cover is important (see Chapter 14). Poorly fitting margins, with or without cement overflow, will almost inevitably result in gingival inflammation. This increases the difficulties associated with taking impressions and fitting of final restorations. It is also true that in susceptible patients periodontal disease may be initiated or show recurrence in such circumstances, with all the ensuing potential problems of treatment at that time. In such cases the aesthetics of the final restorations may also be compromised.

Choice of advanced restorations

In the past, it was often assumed that patients with a positive history of periodontal disease were unsuitable for advanced operative dentistry. This is not true. Provided the periodontal condition has been successfully stabilized and the patient is able to maintain this, provision of well-planned and carefully executed fixed restorative work and/or implants should be considered where justified (Figure 2.12); provision of a removable prosthesis may simply compromise periodontal health.

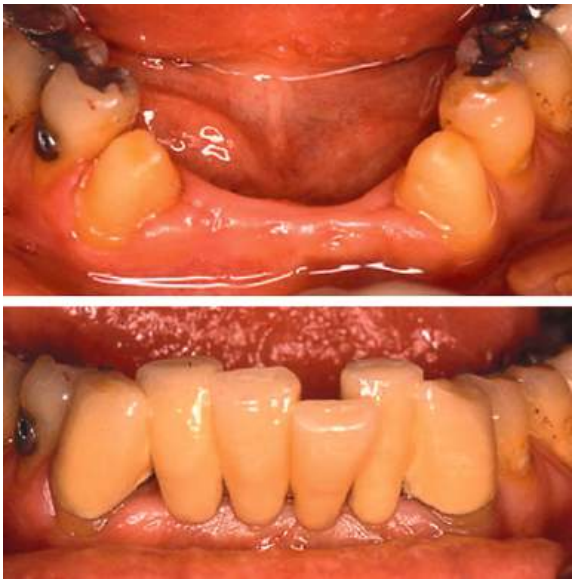


Figure 2.12 Fixed conventional bridge in a patient who has lost lower incisors due to periodontal disease. Successful stabilization of the periodontal disease has enabled fixed prosthodontic treatment. Note supragingival margins on lower canines to ensure maintenance of periodontal health.

Occlusion and the periodontium

If the periodontium is stable, application of even abnormal occlusal forces may cause teeth to become mobile but the changes which occur do not affect the supracrestal periodontal tissues. This means that occlusal trauma alone cannot initiate gingivitis or periodontitis. However, application of normal or abnormal occlusal forces to teeth which already have plaque-induced periodontal disease may result in exacerbation of the periodontal disease.

It must also be remembered that teeth with reduced periodontal support (loss of attachment, bone loss, etc., even if the disease has been controlled and is no longer active) will show signs of occlusal trauma as a result of smaller occlusal loads than would teeth with non-reduced periodontal support. It is essential in these cases in particular to carefully plan and execute restorations which will spread occlusal load efficiently throughout the dentition as much as possible and avoid high pressure on individual elements. Care must also be taken to ensure that there is consideration of the effect of the occlusion in all excursions of the mandible, including right and left lateral and protrusion. This is true at all stages of construction, including temporary coverage.

Signs of occlusal trauma may be clinical or radiographic. Clinically, by far the most important sign is progressive increase in mobility over months or weeks. Other clinical signs include migration of teeth, hypertrophy and hypertonicity of the muscles of mastication, and even signs of temporomandibular disorders (see Chapter 6). Radiographic signs include widening of the periodontal ligament space, funnel-like bone resorption of the crestal bone or loss of definition of the *lamina dura* (Figure 2.13), though this last element should be interpreted with great caution.

If teeth exhibit mobility, occlusal adjustment by spot grinding etc. will be effective in reducing this only if there is widening of the periodontal ligament. If tooth mobility is related to reduced periodontal support and/or periradicular disease, occlusal adjustment will be ineffective.

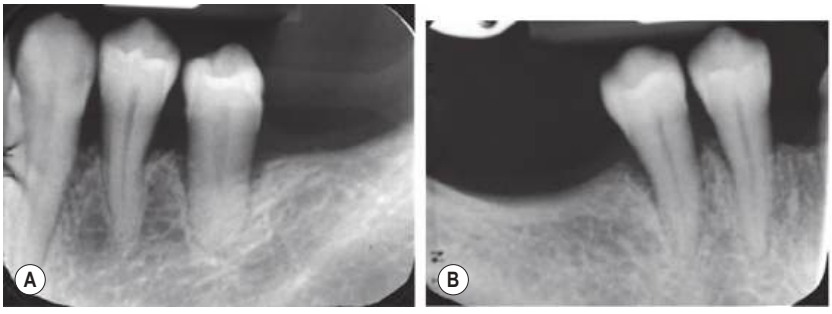


Figure 2.13 Funnel-shaped loss of crestal bone and loss of definition of the lamina dura affecting mandibular premolars as a result of occlusal trauma.

SUMMARY

This chapter has highlighted the importance of establishing periodontal health prior to advanced operative dentistry. To achieve this, a detailed examination, special tests, diagnosis and phased treatment of the periodontium has been discussed. Such a methodical and meticulous approach underpins the success and lays down the foundations for the restorations described in this textbook.

FURTHER READING

Croucher, R., 2005. Why and how to get patients to stop smoking. Dent. Update 32, 143–149.

Tezal, M., Grossi, S.G., Ho, A.W., Genco, R.J., 2004. Alcohol consumption and periodontal disease. The Third National Health and Nutrition Examination Survey. J. Clin. Periodontol. 31, 484–488.

1999 World Workshop system. In: Wiebe, C.B., Putnins, E.E., 2000. The periodontal disease classification system of the American Academy of Periodontology – an update. J. Can. Dent. Assoc. 66, 594–597.

Aggressive periodontitis

Clerehugh, V., 2004. Periodontal diagnosis in young patients. In: Wilson, N.H.F. (Ed.), Periodontal management of children, adolescents and young adults. Quintessence Publishing, London, pp. 86–89.

Hamp, S.E., Nyman, S., Lindhe, J., 1975. Periodontal treatment of multirooted teeth. Results after 5 years. J. Clin. Periodontol. 2, 126–135.